

Project Roles

Anjitha Joby

- Wrote user stories 8, 9, 10
- Created sequence diagrams about Video Editing, Content Recommendations & Cloud Storage
- Created class diagram
- Created UI Mock up 4,5
- Created Test Cases 8,9,10
- Created Test Plan (introduction & approach)
- Created EP + BVA Table section
-

Kieran Hurley

- Wrote User Stories 1,2,3,4
- Created Sequence Diagrams 1,2,3,4
- Created Architecture diagram
- Created Ui Mock Up 2,3
- Created Test Cases for Commenting, Searching, Following and Channel Analytics

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- Wrote User Stories 5,6,7
- Created sequence diagram 5,6,7
- Created Use Case diagram
- Created test cases 5,6,7 and created approach
- Created UI screen 1,6

User Stories

1.As a registered user I want to leave a comment underneath the video I'm currently watching so that I can share my opinion for other users and the creator to see. Clicking the comment button should allow the user to view other comments left by other users, sort comments by 'popular' and 'recent' and vote on other user comments and reply.

2.As a registered user I want to be able to search for specific videos and users so that I can find content that caters to my current needs. Search predictions based on user likes will appear as they type to help speed up the search process, with results sorted by popularity and positive voting.

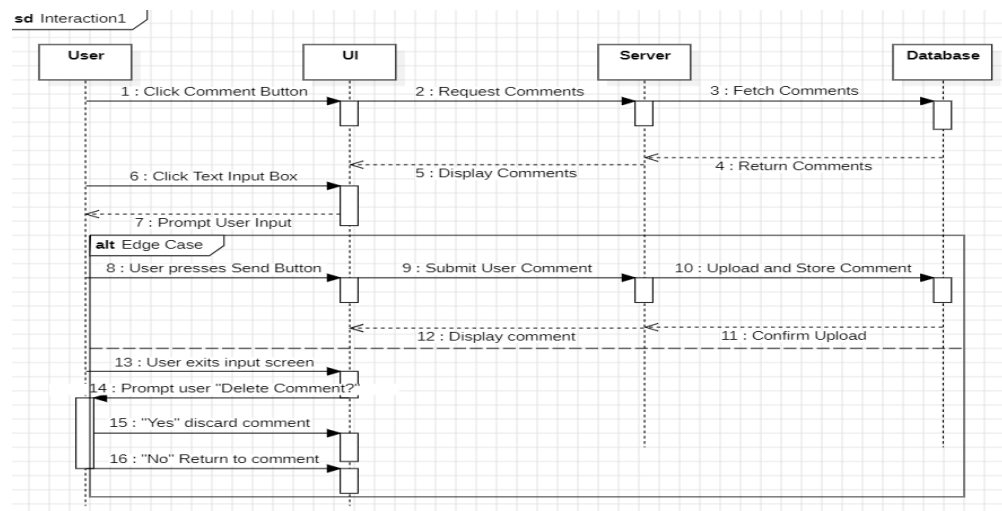
3.As a registered user, I want to be able to follow certain creators so that I can view their uploads and be updated when they make announcements or posts. If the user decides to follow a creator, they will be prompted to enable notifications and once followed, the creator will be recommended more frequently in their feed.

4.As a registered user, I want to be able to view my videos analytics to improve on my content and expand my userbase. Each video will be rated positive or negative to quickly brief the creator on performance, with detailed analytics including views, comments, likes and like-to-dislike ratio accessible via the three-dot button.

5. As a registered user, I want to generate and edit video thumbnails to attract viewers and maintain creative control. This allows users to create and modify video thumbnails, helping them make their content more attractive and maintain creative control.
6. As a registered user, I want to vote on videos and polls to express my preferences and see real-time engagement. This enables users to vote on videos and polls, allowing them to express opinions and see real-time engagement from others.
7. As a registered user, I want to share videos across platforms with proper privacy and tracking to maximize reach and interaction. This feature lets users share videos across different platforms while controlling privacy and tracking interactions to increase reach.
8. As a registered user, I want to be able to create a video where I can import video clips and edit them using different editing tools. This will give me an opportunity to create my own videos by combining and editing clips, allowing me to export a final piece.
9. As a registered user, I want to watch videos based on my watch history, so I can watch content that matches my personal interests. This allows me to watch personalized videos without having to search for them, improving my experience overall.
10. As a registered user, I want to upload videos and drafts to the cloud, so that I can easily access them to prevent losing any work. This will allow me to continue editing, with the opportunity to access my work on different devices.

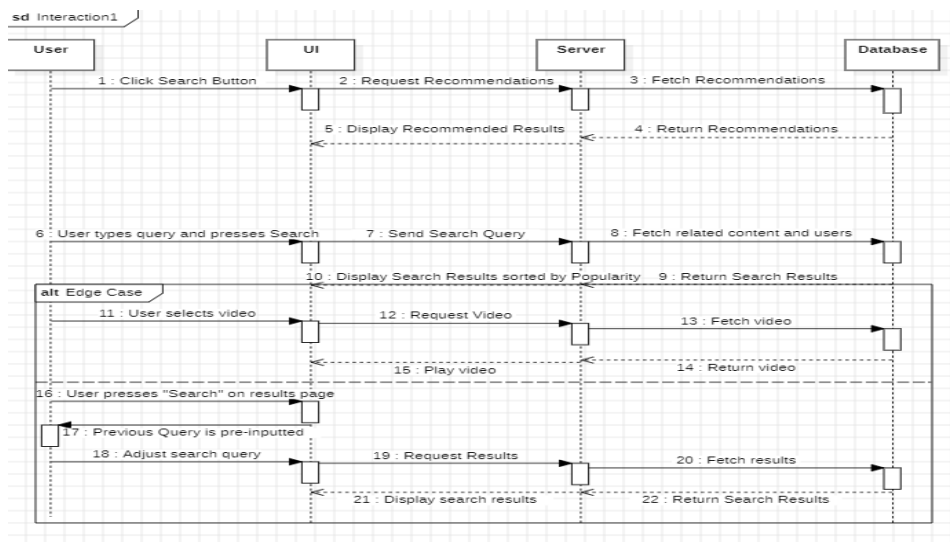
Sequence Diagrams

Comment Diagram:



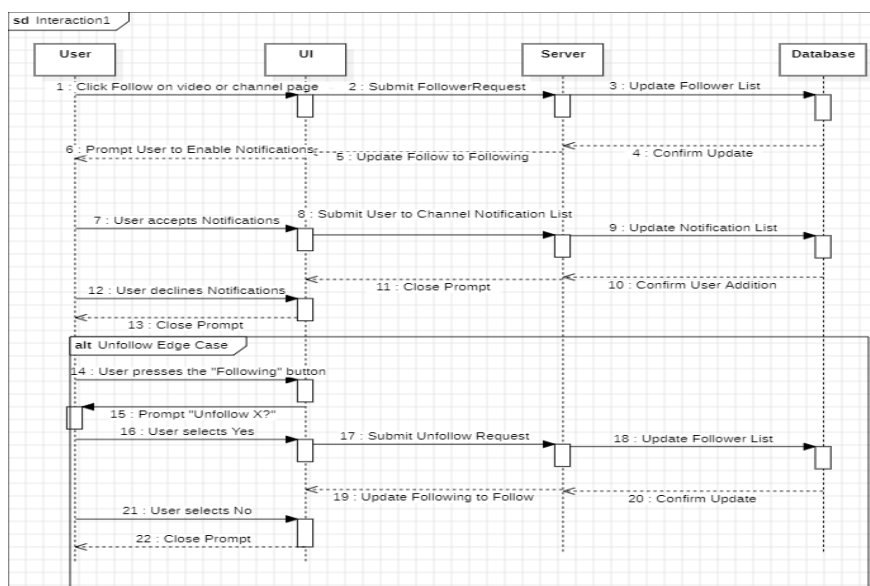
Beginning from the video page, upon pressing the comment icon, we request the comments of the current video from the server and fetch them from the database. Once we return that data, we can show it on the user's display along with a text input box above to allow for commenting. The alt details the commenting process itself, showing the path of a user typing and submitting a comment and an edge case where they exit the input screen mid comment, being prompted to discard their draft or cancel.

Search Diagram



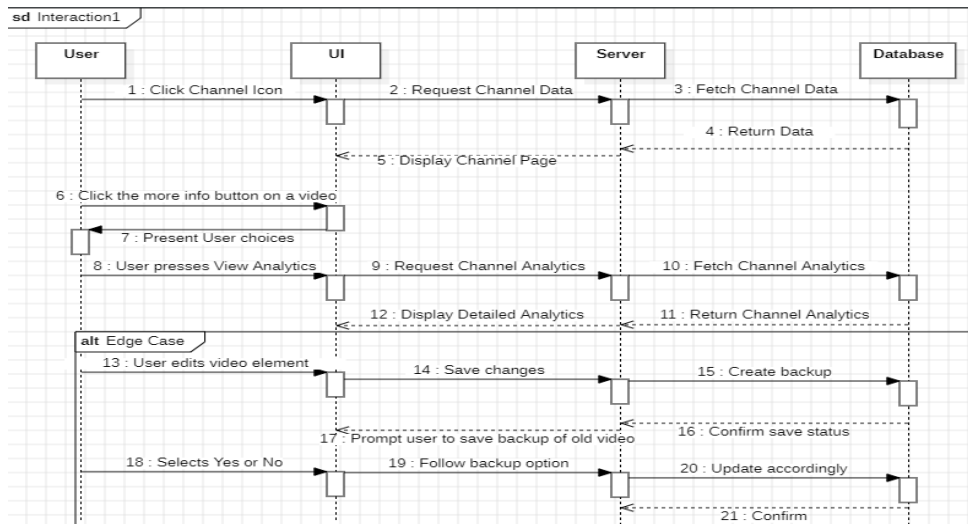
Beginning on the homepage, upon pressing search, search recommendations will be requested from the server, fetched from the database and returned and displayed on the user screen below the search box. Once the user types a search query, it'll be sent to the server, requesting related content and users to the query and return results based on popularity and relevancy. The alt box details two paths, path 1 the user finds a suitable video and chooses to watch it, requesting the necessary data and then playing it once retrieved, whereas the edge case details a user who has not found suitable content and adjusting their previously inputted query and searching once more.

Follow Diagram



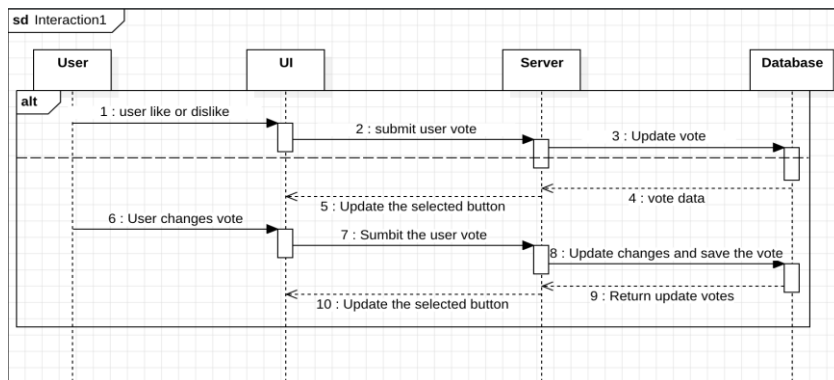
Beginning on either a video or channel page, upon pressing the follow button, we submit a request to the server and update the list of followers on the database before confirming to the user and updating their status, after this they are prompted to enable notifications, if they choose to enable them they are added to a separate list for notifications, if they choose to disable them the prompt is simply closed. Our edge case details what happens when a user is already following the creator, the user is prompted on whether or not they wish to unfollow the creator, if they select "Yes" an unfollow request is submitted and they're removed from any relevant lists and the change is displayed to the user, and if they choose no the prompt is closed.

Video Analytics Diagram:



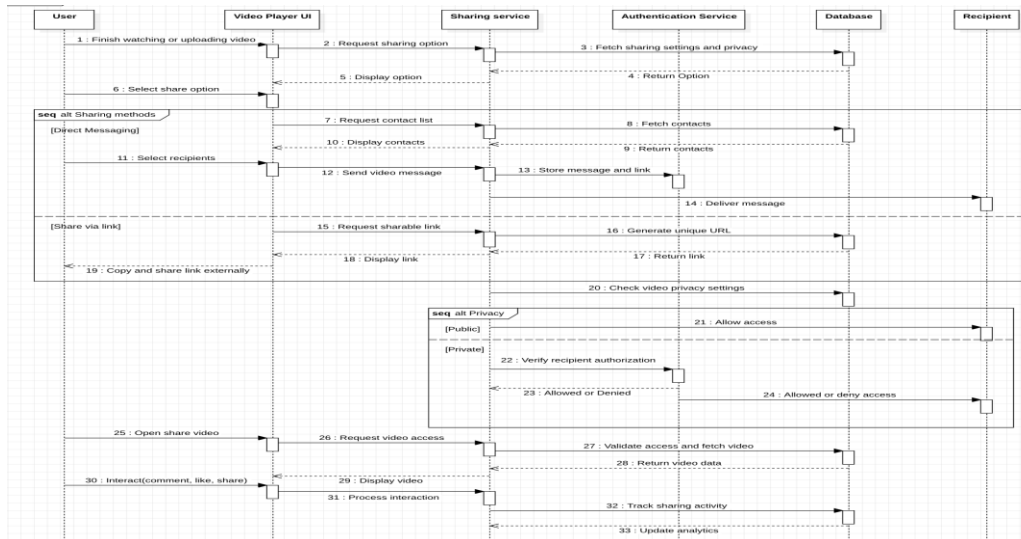
Beginning on either the search, home or video page when a channel icon is pressed, the channel data is requested from the server, fetched, returned and then displayed to the user on channel page. When accessing your own channel page, the videos listed will display brief analytics if available. On pressing the 3 dotted “more” button, a sub menu pops up, presenting the user with choices. Upon selecting the video overview and opening video analytics, the overall analytics will be requested and displayed to the user. On the edge case, once a user makes a change based on our recommendations created by analytics and save, a backup will be created and stored in the database, once the save is confirmed the user will be asked if they wish to save the old video backup, and the backup will be saved accordingly to their choice.

Voting Diagram:



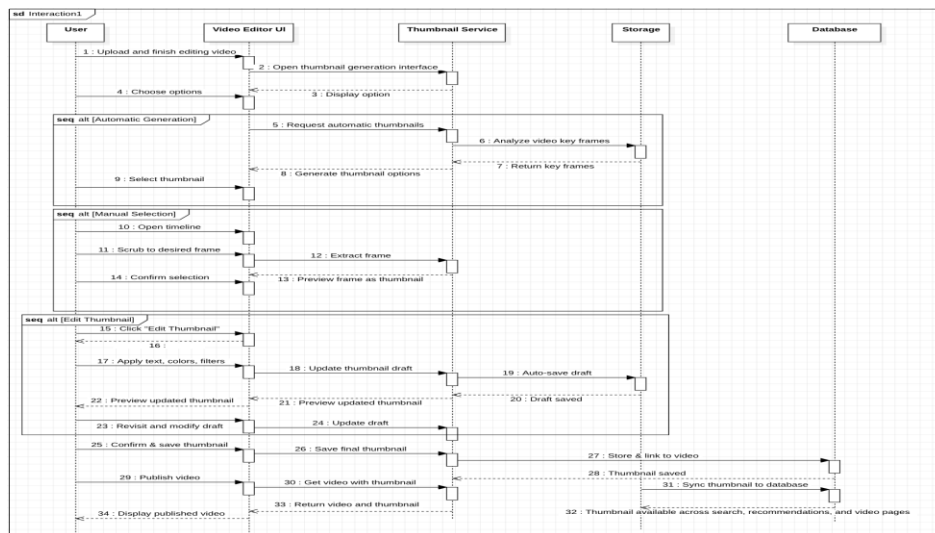
This sequence diagram shows how a voting system processes a user’s like or dislike action. First, the user selects a vote in the UI, which sends the request to the server. The server updates the vote in the database and returns the updated data, allowing the UI to reflect the selected option. If the user changes their vote, the same flow repeats: the updated vote is sent to the server, stored in the database, and the UI is refreshed to show the new selection. Overall, it illustrates the interaction between user, interface, server, and database to keep vote data consistent and up to date.

Video Sharing Diagram:



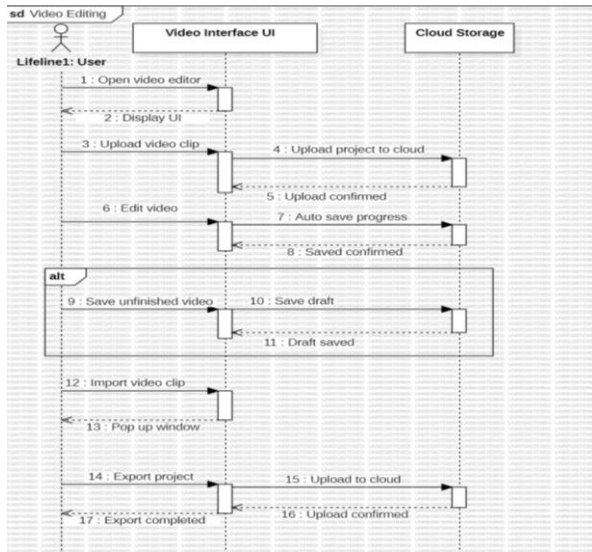
This diagram shows the end-to-end flow of sharing a video, starting when a user selects a sharing option in the video player's UI. The system retrieves sharing settings and allows the user to either send the video directly to contacts or generate a shareable link. For direct sharing, contacts are fetched, recipients are selected, and the message is delivered, for link sharing, a unique URL is created and shared externally. When a recipient accesses the video, the system checks privacy settings, granting access to public content or verifying authorization for private content, then delivering the video. User interactions such as viewing, liking, or commenting are processed and tracked to update analytics.

Thumbnail Generation Diagram:



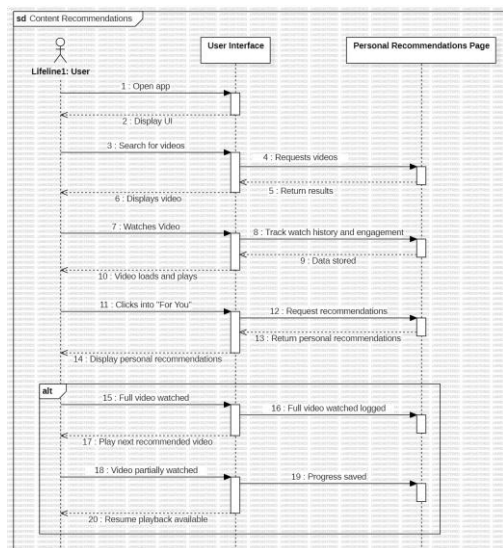
This diagram shows the end-to-end flow of thumbnail generation in a video platform: after a user uploads and edits a video, they access the thumbnail tool and either choose an automatically generated option—based on analyzed key frames—or manually select a frame from the timeline. The chosen thumbnail can be edited with text and visual enhancements, with changes auto saved as drafts. Once finalized, the thumbnail is stored, linked to the video, and synced with the database, making it visible when the video is published and accessible across search, recommendations, and viewing pages.

Video Editing Diagram:



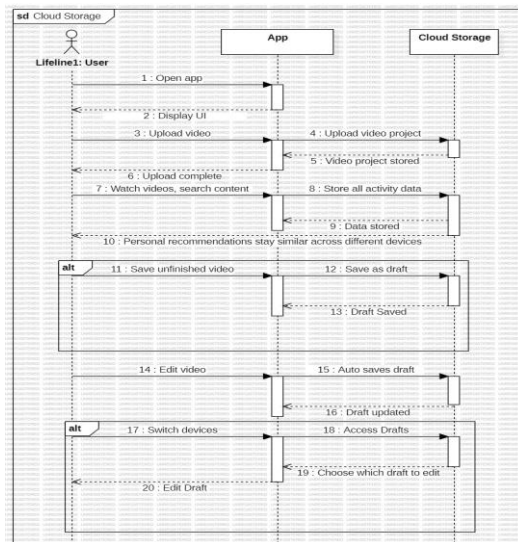
This sequence diagrams describes the video editing feature on the app. When the user opens the app, an editing UI is displayed. The user uploads a clip, from the pop-up window, and the project is stored in cloud. Users can edit the video, while having their progress automatically saved. User also have an option to save their video as a draft if their video is not complete. At the end, users can export their project, which will be automatically uploaded to cloud

Content Recommendations Diagram:



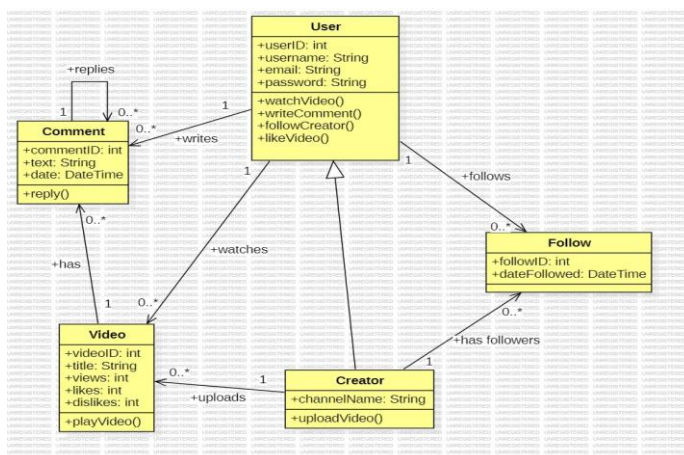
This sequence diagrams describes how the personal recommendations of users are displayed. As the user watches videos, where they are either fully watched or partially watched, all this data is logged. The watch history of the user is stored and then displayed as various similar videos which will appeal to the user.

Cloud Storage Diagram:



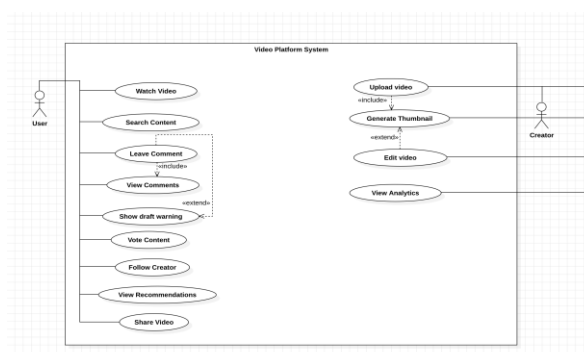
This sequence diagrams describes how the cloud storage works within the app. When user uploads video to the cloud, it is stored preventing any work to be lost. User activity is also recorded and stored in the cloud. When users are editing a video, any edits will be auto saved, and any drafts that users create will be uploaded to cloud. Users will also be able to access their drafts no matter what personal device they use.

Class Diagram:



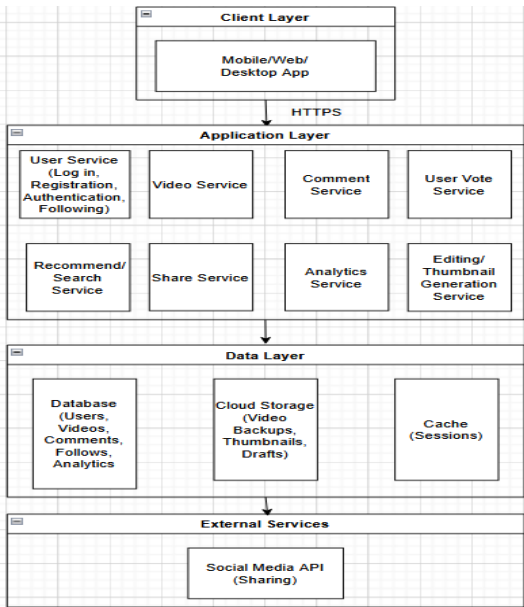
This class diagrams outlines the database for a content creation app. The systems show two different people, users who watches and interacts with videos and content creator who uploads the videos. Users can follow Creators, watch videos and post comments on videos. The video is uploaded by the Creator and the views, likes, dislikes are tracked.

Use Case diagram:



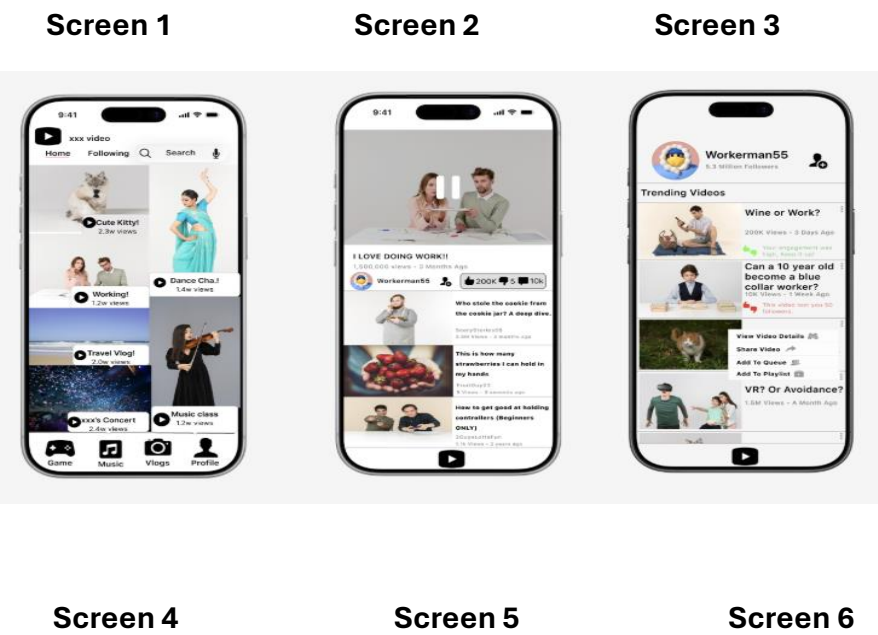
This use case diagram represents the main function of a video platform system for two types, Users and Creators. Users can watch videos, search for content, leave comments, view comments, vote on content, follow creators, view recommendations, and share videos. When a user leaves a comment, the system includes the option to view comments. In some cases, the system may show a draft warning to prevent losing unsaved comments. Creators can upload videos, generate thumbnails, edit videos, and view analytics. The system includes thumbnail generation as part of the upload process, and creators can extend this by editing the video for more customization.

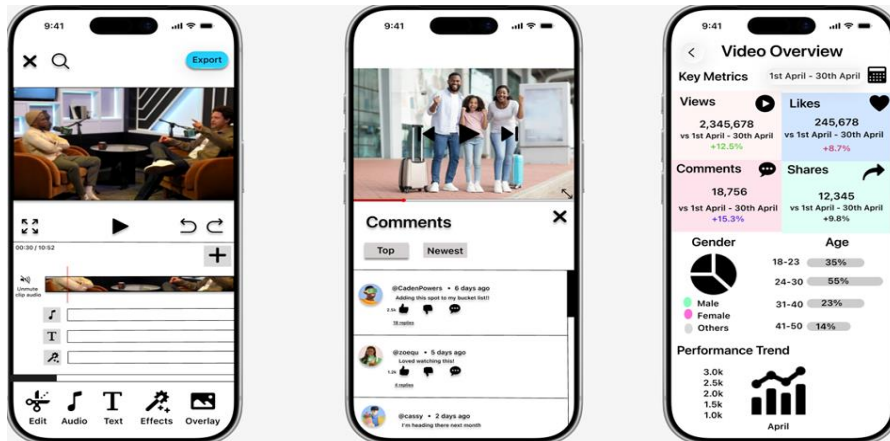
Architecture Diagram:



This diagram represents the high-level architecture for the video platform. Following a layered design that consists of a Client Layer, an application layer consisting of microservices that give functionality (E.g. Comment Service), a data layer consisting of a Database, Cloud storage and Cache and external services ensuring content can be shared off platform. This architecture supports all our function requirements: commenting, voting, video searching, uploading, personalised recommendations, analytics and backups and drafts.

UI Mock Up:





Screen 1: The home screen shows the app's home feed, where users can browse a grid of recommended videos with thumbnails, titles, and view counts. It includes a top navigation bar for sections like Home, Following, and Search, along with a bottom menu for quick access to features such as gaming, music, vlogs, and the user profile.

Screen 2: The video player screen displays a video player at the top of the screen. Below are details of the video such as the title, views, and date posted. Just below and to the left is the channel bar, containing the icon, username and a follow button. To the right of this are buttons allowing for user interactions such as likes, dislikes, and comments relating to the video. The remainder of the screen contains recommendations relating to the user's likes and current video. It contains a navigation bar at the bottom to return to the home page

Screen 3: The channel page screen displays a channel bar containing the username, followers and a follow button. Trending videos of the creator are displayed at the top with a brief analytical overview for each video being displayed, but the analytics are only shown to the creator. Beside each video is a three dotted button that can open a sub menu allowing the user to view video details, share the video, add it to the queue or to a playlist. At the bottom of the page, the navigation bar contains a home button.

Screen 4: The video editor includes the preview window displaying the current frame of the video clip. At the top, there are controls such as the close, search and export button allowing users to exit the project, search content, or finish their project. The editor also contains playback controls (play/pause and undo/redo). The timeline shows the video clip, where at any time users can add text, audio, and effects. The "+" button allows users to add new video clips. The bottom menu shows editing tools users can use. The slider above the bottom menu allows users to slide left or right to use the additional editing tools.

Screen 5: The comment section displays the top users' comments. Users are able to scroll down to view more comments. Users can also interact with the comments by either liking/disliking and adding a response of their own. Each comment posted by a user displays their profile picture and username.

Screen 6: The video overview screen displays a video analytics dashboard, providing insights into performance metrics such as views, likes, comments, and shares over a selected time. It also includes audience demographics (age and gender) and a performance trend chart, helping creators understand how their content is performing and who their viewers are.

System Testing:

Test Plan

Introduction:

This test plan will cover the testing of a video sharing and editing application, which is designed for registered users and creators. The testing will focus on the 10 user stories. This will make sure all requirements are met and all errors are handled.

Scope:**In Scope:**

All user stories will be tested for registered users and creators.

- Commenting on videos
- Searching for videos
- Following creators
- Viewing video analytics
- Thumbnail Editing
- Voting on videos (like/dislike)
- Sharing videos
- Video Editing
- Personalised recommended videos based on watch history
- Uploading videos/drafts to the cloud

Out of Scope:

- Security features
- Creator monetisation features
- Subscription plans
- Copyright
- Payment and billing

Approach:

- The testing approach for this project combines manual, functional, unit, and system testing to ensure all user stories and features work correctly. Both positive and negative test cases are used to validate expected behavior and error handling across features such as commenting, searching, following, analytics, thumbnail generation, voting, sharing, video editing, recommendations, and cloud uploads.
- Equivalence Partitioning (EP) and Boundary Value Analysis (BVA) are applied to input-based features, such as testing comment limits with values like 0, 1, 10,000, and 10,001 characters, to ensure correct handling of valid and invalid inputs. Unit testing focuses on smaller components like input validation and file checks, while system testing verifies complete user workflows. All results are recorded as Pass or Fail, and any issues are retested after fixes to ensure the system meets all requirements and acceptance criteria.

Test Cases**Test Case 1:**

Field	Description
Test Case ID	TC-COMMENT-01
Title	As a User I want to leave a comment
Preconditions	User is already logged in and watching video.
Steps	1. Press Comment button 2. Tap Text Input Box 3. Type and send comment. (With stable internet connection)
Expected Result	Comment is displayed on User's screen and visible for other users
Pass/Fail	

User Story:	"As a User I want to leave a comment"
Acceptance Criteria:	Comment Displayed

TC	STEPS	EXPECTED RESULT
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TC-COMMENT-01	Type and upload comment (With internet)	Comment is displayed on User's screen and visible for other users
TC-COMMENT-02	Type comment and attempt to back out.	User is prompted to confirm they'll discard their drafted comment.
TC-COMMENT-03	Type comment over 10,000 characters	User is over character limit, error is thrown and told to reduce comment size.

Test Case 2:

Field	Description
Test Case ID	TC-SEARCH-01
Title	As a User I want to search for specific videos.
Preconditions	User is already logged in.
Steps	1. Press Search button 2. Tap Text Input Box 3. Type Search Query 4. Confirm search (Query has matching results)
Expected Result	The user is displayed the appropriate content for the query.
Pass/Fail	

User Story:	"As a User, I want to search for a specific video"
Acceptance Criteria:	Search shows matching results "No Results" if nothing matches

TC	STEPS	EXPECTED RESULT
TC-SEARCH-01	Search for a video (Query has valid results)	The user is displayed the appropriate content for the query.
TC- SEARCH -02	Search for a video (Query has no valid results)	The user is told their query yields no valid results.
TC- SEARCH -03	Search for a video and edit query	User's previous query should still be present.

Test Case 3:

Field	Description
Test Case ID	TC-FOLLOW-01
Title	As a User I want to follow my favourite creators.
Preconditions	User is already logged in, User is on either a video or channel page with a follow button present
Steps	1. Press the follow button 2. Select whether to enable notifications
Expected Result	The follow button updates from "Follow" to "Following"
Pass/Fail	

User Story:	"As a User, I want to follow my favourite creators"
Acceptance Criteria:	Button shows "Following" Button shows "Follow" if user chooses to unfollow

TC	STEPS	EXPECTED RESULT
TC- FOLLOW -01	The user presses the "Follow" button. (They are not currently following)	After deciding notification preference, "Follow" should update to "Following"
TC- FOLLOW -02	The user presses the "Follow" button. (They are currently following)	After deciding whether to confirm unfollowing the creator, "Following" should update to "Follow"
TC- FOLLOW -03	The user follows a deactivated creator.	Error message shown, "Follow" stays unchanged

Test Case 4:

Field	Description
Test Case ID	TC-ANALYTICS-01

Title	As a User I want to view my video analytics to make improvements.
Preconditions	User is already logged in, User is on their channel page, User has uploaded videos
Steps	1. Press the 3 dotted button on any video 2. Press view video details 3. Open video overview
Expected Result	An array of video results are presented to the user.
Pass/Fail	

User Story:	"As a User I want to view my video analytics to make improvements."
Acceptance Criteria:	Any form of analytics being displayed Insufficient Data

TC	STEPS	EXPECTED RESULT
TC- ANALYTICS -01	The user views their video and opens their video overview (Video has data)	User should be provided the appropriate analytics
TC- ANALYTICS -02	The user views their video and opens their video overview (Video has insufficient data)	The user should be told their video has insufficient data for analytics.
TC- ANALYTICS -03	The user views their video and opens their video overview (Video has been taken down)	The user should be told that his video no longer exists.

Test Case 5:

Field	Description
Test Case ID	TC-Thumbnail Generation and Editing-01
Title	Generate automatic video thumbnail
Preconditions	User is logged in as a creator and has upload a video
Steps	1.Complete video upload 2. Open thumbnail generation interface 3. Select" Auto-Generate-Thumbnail"
Expected Result	System generates multiple thumbnail options from key frames.
Pass/Fail	

User Story:	"As a register creator, I want to generate and edit video thumbnails"
Acceptance Criteria:	User can generate thumbnails automatically.

TC	STEPS	EXPECTED RESULT
TC- Thumbnail Generation and Editing-02	1.View generates thumbnails 2.Select one thumbnails 3.Confirm selection	Selected thumbnail is saved and applied to video.
TC- Thumbnail Generation and Editing-03	1.Select a thumbnail 2.Edit text/layout/colours 3.Save change	Selected thumbnail is saved and applied to video
TC- Thumbnail Generation and Editing-04(neg)	1. Edit thumbnail 2. Exit without saving	System prompts user to save or discard changes

Test Case 6:

Field	Description
Test Case ID	TC-Voting-01
Title	Vote on a video
Preconditions	User is logged in and watch a video
Steps	Click "like" or "dislike"
Expected Result	Vote is recorded and updated in real time
Pass/Fail	

User Story:	"As a register creator, I want to vote on videos and polls"
Acceptance Criteria:	User can like or dislike the video.

TC	STEPS	EXPECTED RESULT
TC-Voting-01	1. Select poll options 2. submit vote.	Poll results are displayed with updated percentages.
TC-Voting-02	1. Vote on video 2. Attempt to vote again	System prevents multiple voting.
TC- Voting-03	1. Vote on a poll 2. Change selection (if allowed)	Vote updates successfully.

Test Case 7:

Field	Description
Test Case ID	TC-Video Sharing-01
Title	Share video via link
Preconditions	User is logged in and watching a video.
Steps	1. Click "Share" button 2. Select "Copy link" 3. Open the link in browser
Expected Result	Video opens successfully with correct permissions.
Pass/Fail	

User Story:	"As a register creator, I want to share videos across platforms"
Acceptance Criteria:	User can share via link or direct message.

TC	STEPS	EXPECTED RESULT
TC-Video Sharing-01	1. Click share 2. Select user/contact 3. Send	Recipient receives video link.
TC- Video Sharing-02	1. Share private video link 2. Open with unauthorized user	Access is denied.
TC- Video Sharing-03	1. Click share 2. Disconnect internet 3. Attempt to share	Error messages are displayed.

Test Case 8:

Field	Description
Test Case ID	TC-VIDEOEDITOR-01
Title	User wants to import and edit the video
Preconditions	User logged into account, video editor open, video clips imported
Steps	1. Import video clips 3. Apply editing tools 4. Preview the video
Expected Result	Videos successfully imported, editing tools applied, video preview shown
Pass/Fail	

User Story:	As a user, I want to import and edit video clips using different editing tools.
Acceptance Criteria:	Successfully import videos into the video editor

TC	STEPS	EXPECTED RESULT
TC-VIDEOEDITOR-01	User imports and edits the video	Video is imported, edits are reflected in the preview

TC-VIDEOEDITOR-02	User edits without importing a video	Message display "Import video to start"
TC-VIDEOEDITOR-03	User imports unsupported file type	Video not imported, Message displays "Unsupported file format"

Test Case 9:

Field	Description
Test Case ID	TC-RECOMMENDATIONS-01
Title	User wants to see more personal recommendations based on their watch history
Preconditions	User is logged in, user has previously watched videos, watch history recorded
Steps	1.Open app 2. Click into home page 3. Click into a video
Expected Result	User recommended videos displayed based on watch history
Pass/Fail	

User Story:	As a user, I want to watch recommended videos that matches my personal interests.
Acceptance Criteria:	User watch history should be recorded to display user recommendations

TC	STEPS	EXPECTED RESULT
TC-RECOMMENDATIONS-01	Select into home page (watch history exists)	User is shown personalised recommended videos
TC-RECOMMENDATIONS-02	Select into home page (no watch history)	User is shown trending videos.

Test Case 10:

Field	Description
Test Case ID	TC-UPLOAD-01
Title	User wants to upload videos to the cloud
Preconditions	User account exists, system is running, video editor open,
Steps	1.Select "Export" 2. Select "Upload to cloud"
Expected Result	Video successfully uploaded and backed up in cloud. User receives confirmation message.
Pass/Fail	

User Story:	As a user, I want to upload my videos to the cloud.
Acceptance Criteria:	Successfully upload videos to the cloud.

TC	STEPS	EXPECTED RESULT
TC-UPLOAD-01	Select "Export" and then select "Upload to cloud" (With Internet and Storage)	Video successfully backed up in cloud and user receives confirmation message
TC-UPLOAD-02	Select "Export" and then select "Upload to cloud" (With No Internet)	Video backup failed with error message "No internet connection"
TC-UPLOAD-03	Select "Export" and then select "Upload to cloud" (With No Storage)	Video backup failed with error message "No storage available"

EP + BVA Table

Test No.	Test Cases Covered	Inputs	Expected Outputs
1	TC-COMMENT-01	100	True

2	TC-COMMENT-03	11,000	False
3	TC-COMMENT-04*	0	False
4	TC-COMMENT-05*	1	True
5	TC-COMMENT-06*	2	True
6	TC-COMMENT-07*	9,999	True
7	TC-COMMENT-08*	10,000	True
8	TC-COMMENT-09*	10,001	False

Scenario	Test Step	Expected Result	Actual Outcome	Level
Verify that the input field accepts comment within limit	Enter 100 characters and upload	Application should be able to accept all 100 characters comment	Application accepts all 100 characters	System
Verify that the input field rejects comment over the limit	Enter 11,000 characters and upload	Application should not accept all 11,000 characters comment and show error	Application rejects comment, shows error to reduce comment size	System
Verify that the input field rejects empty field	Enter 0 characters and upload	Application should reject empty comment	Application rejects empty comment, shows error	Unit
Verify that the input field accepts comment of minimum of 1 character	Enter 1 character and upload	Application should be able to accept 1 character comment	Application accepts all 1 character	Unit
Verify that the input field accepts comment just above minimum of 2 characters	Enter 2 characters and upload	Application should be able to accept all 2 characters comment	Application accepts all 2 characters	Unit
Verify that the input field accepts comment just below maximum of 9,999 characters	Enter 9,999 characters and upload	Application should be able to accept all 9,999 characters comment	Application accepts all 9,999 characters	Unit
Verify that the input field accepts comment with maximum of 10,000 characters	Enter 10,000 characters and upload	Application should be able to accept all 10,000 characters comment	Application accepts all 10,000 characters	Unit
Verify that the input field rejects comment over maximum of 10,001 characters	Enter 10,001 characters and upload	Application should not accept all 10,001 characters comment and show error	Application rejects comment, shows error to reduce comment size	Unit